

Question			Answer/Indicative content	Marks	Guidance
1			A	1	
			Total	1	
2			B	1	
			Total	1	
3			A	1	
			Total	1	
4	a		The emission of electrons from the surface of a metal when electromagnetic waves (of frequency greater than the threshold frequency) are incident on the metal. The wave model cannot explain why there	B1	
	b		is a threshold frequency for metals. The new model / photon model proposed one-to-one interaction between photons and electrons and this successfully explained why threshold frequency exists. Any further one from: Energy of photon (hf) must be greater than or equal to work function of metal. The kinetic energy of emitted electrons was independent of the incident intensity. Correct reference to $hf = \Phi + K_{E_{max}}$	B1 B1 B1	
	c	i	$E = \frac{6.63 \times 10^{-34} \times 3.0 \times 10^8}{380 \times 10^{-9}}$ or $\phi = 1.1 \times 1.6 \times 10^{-19}$	C1	This is substituting values into $\frac{hc}{\lambda} = \phi + \frac{1}{2}mv^2$
		i	$\frac{6.63 \times 10^{-34} \times 3.0 \times 10^8}{380 \times 10^{-9}} = 1.1 \times 1.6 \times 10^{-19} + \frac{1}{2} \times 9.11 \times 10^{-31} v^2$	C1	
		i	speed = 8.7×10^5 (m s ⁻¹)	A1	
		ii	The energy of a photon depends only on wavelength or frequency, so intensity does not change the maximum speed of the photoelectrons.	B1	
			Total	8	

Question			Answer/Indicative content	Marks	Guidance
5			energy of <u>two</u> photons = $2 \times mc^2$ or $2 \times \frac{hc}{\lambda} = 2 \times mc^2$	C1	
			$\lambda = \frac{h}{mc} = \frac{6.63 \times 10^{-34}}{9.11 \times 10^{-31} \times 3.0 \times 10^8}$		
				C1	Correct use of $\frac{hc}{\lambda} = mc^2$
			wavelength = 2.4×10^{-12} (m)	A1	
			Total	3	
6			D	1	
			Total	1	
7			A	1	
			Total	1	
8	a		$\frac{hc}{\lambda} = \phi + KE_{\max}$ and $\phi = 2.3 \times 1.6 \times 10^{-19}$	C1	
			$\frac{6.63 \times 10^{-34} \times 3.00 \times 10^8}{5.1 \times 10^{-7}} = 2.3 \times 1.6 \times 10^{-19} + KE_{\max}$	C1	
			$KE_{\max} = 2.2 \times 10^{-20}$ (J)	A1	Allow 3 marks for an answer of 2.0×10^{-20} J; value of h to 2 s.f. is used.
	b		The rate of photons incident on M is doubled.	B1	
			The rate of emission of photoelectrons / current is doubled.	B1	
			Total	5	
9			$\frac{hc}{\lambda} = 2 \times 9.11 \times 10^{-31} \times (3.0 \times 10^8)^2$	C1	
			$\lambda = \frac{6.63 \times 10^{-34} \times 3.0 \times 10^8}{2 \times 9.11 \times 10^{-31} \times (3.0 \times 10^8)^2}$	C1	
			$\lambda = 1.2 \times 10^{-12}$ (m)	A1	Allow 2 marks for 2.4×10^{-12} (m); factor of 2 omitted in the first line.
			Total	3	

Question			Answer/Indicative content	Marks	Guidance
10		i	Fission reactors produce radioactive by-products which affect future generations and the environment in terms of possible contamination / exposure to humans and animals.	B1	
		ii	No of particles in 1000 g U = $1000/235 \times 6.02 \times 10^{23} = 2.56 \times 10^{24}$ No of reactions for U = 2.56×10^{24}	B1	Appreciate that the key to the answer is the difference in numbers of atoms / nuclei or equal number of nucleons involved scores one mark if nothing else achieved.
		ii	Energy from U = $2.56 \times 10^{24} \times 200 = 5.12 \times 10^{26}$ MeV	B1	
		ii	No of particles in 1000g H = 6.02×10^{26} No of reactions = $6.02 \times 10^{26}/4$ Energy from H = $6.02 \times 10^{26}/4 \times 28 = 42.14 \times 10^{26}$ MeV	B1	
		ii	Hence energy $42/5 = 8.2$ times higher	B1	
		ii	<i>second method</i> 235 g of U and 4 g of H / He contain 1 mole of atoms	or B1	
		ii	there are 4.26 moles of U and 250 moles of He	B1	
		ii	so at least 58 times as many energy releases in fusion ratio of energies is only 7 fold in favour of U	B1	
		ii	therefore 58/7 times as much energy released by 1 kg of H	B1	
		ii	<i>similar alternative argument, e.g.</i> For U each nucleon 'provides' 0.85 MeV	B1	
		ii	For H each nucleon 'provides' 7 MeV	B1	
		ii	(Approx) same number of nucleons per kg of U or H	B1	

Question			Answer/Indicative content	Marks	Guidance
		ii	so 8.2 times as much energy from H	B1	
			Total	5	



EXAM PAPERS PRACTICE

Question			Answer/Indicative content	Marks	Guidance
11	a		5.0 eV = 8.0×10^{-19} (J) and 2.0 eV = 3.2×10^{-19} (J)	B1	Allow correct answers in terms of threshold frequency / wavelength for the metals and the frequency / wavelength of the photon
			$\text{photon energy} = \frac{6.63 \times 10^{-34} \times 3.0 \times 10^8}{300 \times 10^{-9}} = 6.6(3) \times 10^{-19} \text{ (J)}$	B1	Allow first two B1 marks for photon energy quoted as 6.6×10^{-19} J and 4.1 eV
			energy of photon > work function of X Or energy of photon < work function of Y	B1	
			Hence electrons emitted from X with speed / KE from zero to a maximum value and no electrons are emitted from Y	B1	
	b		*Level 3 (5–6 marks) Clear explanation and discussion <i>There is a well-developed line of reasoning which is clear and logically structured. The information presented is relevant and substantiated.</i> Level 2 (3–4 marks) Some explanation and some discussion <i>There is a line of reasoning presented with some structure. The information presented is in the most-part relevant and supported by some evidence.</i> Level 1 (1–2 marks) Limited explanation or limited discussion <i>The information is basic and communicated in an unstructured way. The information is supported by limited evidence and the relationship to the evidence may not be clear.</i> 0 marks No response or no response worthy of credit. Total	B1 × 6	Indicative scientific points may include: Explanation • $hf = \Phi + K_{\text{Emax}}$ (any subject) • A graph of K_{Emax} against f is a straight line graph with gradient = h (and intercept = $-\Phi$) • Draw a straight best-fit line through points and determine the gradient using a 'large triangle' Discussion of accuracy and precision • % uncertainties are 4.8% for A and 9.1% for B • Data points widely spread out for B. (ORA) • For B the value of h is accurate because its closer to the real / actual value (but the results are not precise) • For A the value of h is precise because of the smaller % uncertainty (but the result is not accurate)
				10	

Question			Answer/Indicative content	Marks	Guidance
12		i	The direction of the electric field due to the negative charge is to the left and right for the positive charge.	B1	
		i	The magnitude of the electric field strength due to the positive charge is smaller than that for the negative charge (because of greater distance). (Hence the resultant electric field strength is to the left.)	B1	
		ii	$\text{energy} = \frac{Qq}{4\pi\epsilon_0 r} = \frac{(1.60 \times 10^{-19})^2}{4\pi\epsilon_0 \times 3.0 \times 10^{-10}}$	C1	
		ii	energy = 7.67(2) × 10 ⁻¹⁹ (J)	C1	
		ii	energy = 4.8 (eV)	A1	
			Total	5	
13		i	E = hc/λ; Δε = E ₁ – E ₂ = hcΔλ/λ ²	C1	allow calculation of E = hc/λ twice and difference taken
		i	Δε = 6.63 × 10 ⁻³⁴ × 3 × 10 ⁸ × 0.6 × 10 ⁻⁹ / 5.92 × 10 ⁻¹⁴	C1	
		i	Δε = 3.4 × 10 ⁻²² (J)	A1	
		ii	sin θ = nλ/d; 1/d = 3 × 10 ⁵ (m ⁻¹)	C1	or similar allow 20.72 – 20.70
		ii	θ ₁ – θ ₂ = sin ⁻¹ (2 × 589.6 × 3 × 10 ⁻⁴) – sin ⁻¹ (2 × 589 × 3 × 10 ⁻⁴)	M1	
		ii	θ ₁ – θ ₂ = 20.717 – 20.695 = 0.0220	A1	
			Total	6	

Question			Answer/Indicative content	Marks	Guidance
14	a		$E = \frac{6.63 \times 10^{-34} \times 3.0 \times 10^8}{1.1 \times 10^{-6}}$	B1	Values must be substituted
			$E = 1.8 \times 10^{-19} \text{ (J)}$	A0	Answer to 3sf is $1.81 \times 10^{-19} \text{ (J)}$ Examiner's Comments This question was specifically included to give a hint as to the method to be used in (b). The question was written in a 'show' format to enable candidates to answer (b) even if they could not recall this area of synoptic work. However this did mean that all working, including substitution, had to be shown and this did result in a small number losing the mark.
	b		$m = \rho V = 8.1 \times 10^{-12} \times 4.5 \times 10^3 = (3.645 \times 10^8)$	C1	Allow: ecf from (a) and mass of titanium Examiner's Comments Again this question had three distinct strands to the physics. The vast majority of candidates were capable of determining the correct mass and thermal energy required to raise the temperature of the titanium. A small number of errors were seen in these two strands however: mainly in transposition of the density formula and converting temperature changes incorrectly to kelvin scale. The final stage to determine the time was less confidently handled with transposition errors and some strange manipulation of the equations which usually resulted in the reciprocal of the correct answer. Perhaps the very small time involved in this form of welding surprised a few candidates.
			Thermal energy gained = $(mc \Delta\theta) = 3.645 \times 10^{-8} \times 520 \times [1700 - 20] (= 0.0318)$ $1.81 \times 10^{-19} \times 6.3 \times 10^{19} \times t = 0.0318$	C1	
			$t = 2.8 \times 10^{-3} \text{ (s)}$	A1	
	c		Thermal energy is conducted / transferred to the rest of titanium / metal	B1	Not: heat lost to surroundings

Question			Answer/Indicative content	Marks	Guidance
			Photons are reflected / scattered from / not absorbed the titanium surface	B1	Examiner's Comments The answers given for this question were disappointing. All too often the only factor quoted was the vague '<i>heat lost to the surroundings</i>'. A significant number of candidates scored one mark by identifying the loss of thermal energy to the non-shaded volume of titanium. Only a tiny minority realised that some photons would be reflected from the metal surface. Other suggestions such as '<i>photons are absorbed in the air</i>', '<i>photons would miss the target</i>', '<i>not all photons have the same energy</i>', '<i>the laser needs to heat up as well</i>' were not given any credit. Marks for this discriminating question were mostly awarded only to the more able candidates. Allow: energy is used to overcome the
	d		(Photon) energy is converted into potential energy (rather than kinetic energy) OR Energy is used to change solid to liquid / phase (rather than increase kinetic energy) OR Energy provides (specific) latent heat of fusion (rather than increase kinetic energy)	B1	forces between atoms / breakdown the crystal structure of titanium (rather than increase kinetic energy) Examiner's Comments This question discriminated across the entire spectrum of ability, largely as a result of candidates writing about the lack of a temperature change rather than focusing on what actually happened to the energy supplied at this stage. Many answers were merely statements lacking in the vital explanation. It was, however, encouraging to see that the physics involved in this unfamiliar situation was broadly understood by the candidates.
			Total	7	
15			D	1	
			Total	1	
16			D	1	
			Total	1	

Question		Answer/Indicative content	Marks	Guidance
17	a	Photon(s) mentioned One-to-one interaction between photons and electrons Energy of photon is independent of intensity / intensity is to do with rate (of photons / photoelectric emission) / photon energy depends on frequency / energy of photon depends on wavelength / photon energy $\propto$ frequency / photon energy $\propto 1/\lambda$ energy of uv photon(s) > work function (of zinc) / frequency of uv > threshold frequency	B1 B1 B1 B1	Allow 'photon absorbed by an electron' Allow: collide etc. for interaction Allow $E = hf$ or $E = hc/\lambda$ Allow energy of light photon(s) < work function (of zinc) / frequency of light > threshold frequency Allow $\geq$ instead of > here Not $f > f_0$ Examiner's Comment Many candidates wrote enthusiastically about photoelectric effect and understood the significance of work function energy (or threshold frequency) and the one-to-one interaction between photon and an electron. Some candidates did not mention 'photons' and this limited the marks they could acquire. The role of intensity was less understood. Many candidates thought it was linked to 'the <i>number of photons</i>' or 'the <i>amount of electrons emitted</i>'. The important term rate of the missing ingredient. Top-end candidates gave eloquent answers, typified by the response: '<i>intensity of visible light only affects the rate of photons incident on the plate but not the energy of each photon</i>'. Two common misconceptions were: • Photons were emitted from the negative plate. • Confusing threshold frequency and work function energy.

Question		Answer/Indicative content	Marks	Guidance
	b	$\phi = \frac{6.63 \times 10^{-34} \times 3.0 \times 10^8}{2.9 \times 10^{-7}} \text{ or } 6.86 \times 10^{-19} \text{ (J)}$ $E = 5.1 \times 1.60 \times 10^{-19} \text{ or } 8.16 \times 10^{-19} \text{ (J)}$ $\text{max kinetic energy} = (8.16 - 6.86) \times 10^{-19}$ $\text{max kinetic energy} = 1.3 \times 10^{-19} \text{ (J)}$	C1 C1 A 1	Note: Using 5.1 and not 8.16×10^{-19} cannot score this mark or the next mark Allow 2 marks for 0.81 eV Examiner's Comment This was a notable success for most of the candidates. Examiners were pleased to see a range of techniques being used to get the correct answer of 1.3×10^{-19} J. Many answers showed excellent structure, effortless conversion of energy from electronvolt to joule and excellent use of the calculator when dealing with powers of ten. Most candidates scored three marks. A small number of candidates left the final answer as 0.81 eV; the only thing missing was the conversion to J.

Question			Answer/Indicative content	Marks	Guidance
	c		Any <u>three</u> from: The electrons are repelled by C / electrons travel against the electric field (AW) The electrons are emitted with a 'range' of speed / velocity / kinetic energy (AW) As V increases the slow(er) electrons do not reach C and hence / decreases maximum KE in the range 2.1 eV to 2.2 eV or 3.36×10^{-19} J to 3.52×10^{-19} J	B1*3	Note 'range' can be implied by 'highest' or 'lowest' Allow 'find p.d. when current is (just) zero, and then $KE = e \times V$ Examiner's Comment The electrons emitted from the metal plate have a range of kinetic energy. The emitted electrons are repelled by the negative electrode C. Fewer electrons reach C as the p.d. is increased. When the p.d. is about 2.2 V, and the current zero, the most energetic electron are stopped from reaching C. This makes the maximum kinetic energy of the electrons equal to 2.2 eV or 3.4×10^{-19} J. The question baffled most candidates. Some top-end candidates commented on '<i>the electrons repelled by C</i>' and the maximum kinetic energy of the emitted electrons being 2.2 eV. Such answers were rare. Too many candidates made guesses with answers such as '<i>the current drops because resistance increases</i>' and '<i>temperature increases and hence the current decreases</i>'.
			Total	10	

EXAM PAPERS PRACTICE

Question			Answer/Indicative content	Marks	Guidance
18		i	<u>electron</u> bound to nucleus / represents energy <u>electron</u> must gain to leave the atom / total energy of <u>electron</u> in atom is less than that of a free electron	B1	Allow ionisation level defined as zero as AW for 'represents electron must gain energy to leave atom / move up energy level' Allow potentials for attractive forces are negative. Examiner's Comments This item provided good discrimination between the candidates. Many responses referred incompletely to the negative charge of the electron being the only factor, whereas the correct explanation is much more to do with the electron requiring energy to leave the atom and the ionization level being defined as the zero point. Some candidates were on the right path when they referred to the equivalent statement for gravitational potential energies. Ignore sign Possible ECF from (ii)1
		ii	1 energy = 2.55 (eV) 2 energy = $2.55 \times 1.60 \times 10^{-19}$ (J) $\lambda = \frac{6.63 \times 10^{-34} \times 3.0 \times 10^8}{2.55 \times 1.60 \times 10^{-19}}$ (Allow any subject) wavelength = 4.9×10^{-7} (m) wavelength = 490 (nm)	B1 C1 C1 A1	Note: wavelength = 488 (nm) to 3 sf Examiner's Comments Virtually all candidates correctly evaluated the energy difference to be 2.55 eV. Negative values were condoned but are unlikely to be accepted in future series. Many candidates correctly calculated the wavelength of emitted light, although a minority did not convert the energy into joules or performed the required conversion to nanometres incorrectly.
			Total	5	

Question		Answer/Indicative content	Marks	Guidance
19	a	${}^{238}_{92}\text{U} \rightarrow {}^{234}_{90}\text{Th} + \dots$ ${}^4_2\text{He}$ or ${}^4_2\alpha$	B1 B1	allow proton and / or nucleon number to the right of symbol allow γ -photon; zero for any other extra particle Examiner's Comments Most candidates made a good start to the paper writing a correct equation for the nuclear decay. allow 0.07×10^{-25} for α -particle mass
	b	$mv = (4.00 - 0.0665) \times 10^{-25} \times 2.40 \times 10^5$ $= 9.44 \times 10^{-20}$ $v = 9.44 \times 10^{-20} / 6.65 \times 10^{-27} = 1.42 \times 10^7$ $\text{k.e.} = \frac{1}{2} \times 6.65 \times 10^{-27} \times (1.42 \times 10^7)^2$ $= 6.70 \times 10^{-13} \text{ (J)}$ $6.70 \times 10^{-13} / 1.60 \times 10^{-13} = 4.19 \text{ (MeV)}$	C1 C1 A1 B1	max 3 if use 4.00 instead of 3.93 in momentum eq'n allow ratio of masses 234 and 4 or calculations using $234u$ and $4u$ allow $p^2/2m$ calculation for k.e. accept 4.0 to 4.2; ecf (calculated value of k.e. in J)/e N.B. the correct answer automatically gains all 4 marks Examiner's Comments One mark in this question was reserved for converting units from joule into mega electronvolt. This was the only mark awarded to half of the candidates. Few recognised this to be an isolated system, applying the conservation of momentum to solve the problem. Few appeared to realise that the mass of an alpha particle is given in the Data, Formulae, and Relationships Booklet, calculating it instead by summing the masses of neutrons and protons. The most common incorrect approach was to use the formula $E = mc^2$ or to equate the kinetic energies of the thorium nucleus and alpha particle.

Question			Answer/Indicative content	Marks	Guidance
	c		$\Delta A = 32 = 4n_{\alpha}$ so $n_{\alpha} = 8$ $\Delta Z = 10 = 2n_{\alpha} - n_{\beta}$ so $n_{\beta} = 6$ argument / reasoning given for both n_{α} and n_{β}	B1 B1 B1	allow 8 (decays), i.e no mention of α particles allow $10 - 16 = -6$; NOT $14 - 8 = 6$; must state $\beta(-)$ particles e.g. change in mass number caused by α decay, change in proton number combination of α and β Examiner's Comments A significant number had no idea where to start and left the page blank. Of the rest most managed to decide on 8 alpha particles. A minority worked initially with the proton number rather than the nucleon number incorrectly choosing 5. The explanations about the choice of 6 beta particles were often just restricted to equating the numbers correctly rather than giving any description of the transformation of neutrons into protons.
			Total	9	

Question			Answer/Indicative content	Marks	Guidance
20	a		$E = (hc/\lambda) = 6.63 \times 10^{-34} \times 3.0 \times 10^8 / 450 \times 10^{-9}$	C1	
			$E = 4.42 \times 10^{-19} \text{ (J)}$	C1	
			energy = 2.76 (eV)	A1	N.B. the answer here must be 2 SF or more
	b	i	$2.76 - 2.3 = 0.46 \text{ eV}$ (so only 0.5% of energy/AW)	B1	allow $2.8 - 2.3 = 0.5 \text{ eV}$ and $3.0 - 2.3 = 0.7 \text{ eV}$ possible ecf from (b)
		ii	$n = 2000 \times 49 (= 5.24 \times 10^8)$ $Q = ne = 8.4 \times 10^{-11} \text{ (C)}$ $I = 8.4 \times 10^{-11} / 2.5 \times 10^{-9}$ average current = 0.034 (A)	C1 C1 A1	allow ecf for wrong n allow 34 m(A); answer is $1.7 \times 10^{-5} \text{ A}$ if 2000 omitted (2/3) Examiner's Comments Almost all of the candidates attempted this last section of the paper with some success. In part (i) most candidates showed that they understood the theory behind the question and subtracted the appropriate two numbers from part (b) to gain the mark. Part (ii) was done well with a significant number obtaining the correct answer. Another large group forgot that 2000 electrons were released and performed the calculation for only a single electron being multiplied up and so forfeited the final mark.
			Total	7	
21			C	1	
			Total	1	

Question	Answer/Indicative content	Marks	Guidance
22	Any <u>one</u> from: Energy of visible light photon < work function (of zinc) (frequency of) visible (light/photon) < threshold frequency Any <u>one</u> from: Energy of UV photon > work function (of zinc) (frequency of) UV (radiation/photon) > threshold frequency Any two from: — • Collapse of leaf linked to removal of electrons • One-to-one interaction of photon and (surface) electron • Photon energy is independent of intensity / Intensity linked to rate of photons (incident on the zinc plate)	B1 B1 B1 × 2	Allow f for frequency, λ for wavelength and ϕ for work function throughout Allow 'overcome' / 'met' / 'reached' when describing > or < Allow photons — Not f_0 for threshold frequency Allow equivalent statement with wavelength Allow = instead of > or < throughout for UV Allow equivalent statement with wavelength Ignore stem / plate / leaf / electroscope becoming positive <u>Examiner's Comments</u> This question on the photoelectric effect was enthusiastically answered by candidates. Good discrimination enabled many of the top-end candidates to score full marks. The question was scrutinised well with many candidates explaining why the leaf fell with the ultraviolet radiation. The one-to-one interaction between UV photons and surface electrons was nicely embedded in the descriptions. Some of the candidates would have benefitted by writing their answers in bullet points.  There were a few missed opportunities and misconceptions, these are outlined below:

Question			Answer/Indicative content	Marks	Guidance
					• Photons were emitted from the zinc plate (rather than electrons) • Threshold frequency and work function were properties of the photons or electrons (and not zinc) • Threshold frequency and work function were synonymous • Intensity was linked to number of photons (rather than to the rate of photons) • Referring to '<i>not enough energy</i>' instead of work function of the metal in the description
			Total	4	



EXAM PAPERS PRACTICE

Question			Answer/Indicative content	Marks	Guidance
23		i	Kinetic energy (of proton) changes to potential (energy) or Potential energy increases as the kinetic energy (of the proton) decreases or Potential energy increases as work is done against the field / against repulsion / positive charge	B1	Allow 'it' / PE for (electric) potential energy Allow KE / E_k

Question			Answer/Indicative content	Marks	Guidance
		ii	energy = $0.52 \times 106 \times 1.60 \times 10^{-19}$ or $8.3(2) \times 10^{-14}$ (J) $\frac{1.60 \times 10^{-19} \times 27 \times 1.60 \times 10^{-19}}{4\pi\epsilon_0 R} = 8.32 \times 10^{-14}$ $R = 7.5 \times 10^{-14}$ (m)	C1 C1 A1	Allow 2 mark for 1.6×10^{-13} (m); Z = 59 used Allow 2 mark for 8.9×10^{-14} (m); Z = 32 used Allow 1 mark for 2.8×10^{-15} (m); Z = 1 used Allow 1 mark for 1.2×10^{-32} (m); energy = 5.2×10^5 used Examiner's Comments The above question on electric potential energy provided excellent discrimination with middle and upper quartile candidates showing how to produce immaculate answers — identify the physics, write down the correct physical equation, do any necessary conversions (e.g. MeV to J), rearrange the equation, substitute correctly and then write the final answer in standard form to the correct number of significant figures. About a third of the candidates scored full marks. ? Some of the missed opportunities or errors were: - Using an incorrect equation with the distance squared - Not correctly converting the kinetic energy 0.52 MeV into joule (J) - Using the equation $r = r_0 A^{1/3}$ for the mean radius of a nucleus to determine the minimum distance
			Total	4	

Question			Answer/Indicative content	Marks	Guidance
24		i	Proton is repelled (by nucleus)	B1	Allow 'proton can experience the strong (nuclear) force' Not 'collide / hit nucleus'
			(High-speed) proton can get close to (oxygen) nucleus	B1	
		ii	$E = [0.25 - (2.24 - 2.20)] \times 10^{-11} \text{ (J) or } 0.21 \times 10^{-11} \text{ (J)}$	C1	Allow 2 marks for 6.9×10^{-14} ; $E = 0.29 \times 10^{-11}$ used Allow 1 mark for a value correctly calculated based on any other incorrect value for E (e.g. $8(.0) \times 10^{-14}$ for $E = 0.25 \times 10^{-11}$ and $5(.0) \times 10^{-13}$ for $E = 0.04 \times 10^{-11}$)
			$\lambda = \frac{6.63 \times 10^{-34} \times 3.00 \times 10^8}{0.21 \times 10^{-11}}$ (Any subject)	C1	
			$\lambda = 9.5 \times 10^{-14} \text{ (m)}$	A1	
		iii	Used in PET (scans)	M1	Enter text here.
			Any <u>one</u> from: Used to diagnose function of organ / brain / body Detection of cancer / tumour Non-invasive / no surgery / no infection 3D (image)	A1	
			Total	7	

Question			Answer/Indicative content	Marks	Guidance
25		i	$(P =) 0.01 \times 2.5$ or 0.012×250 or $2.52/250$ or 0.025 (W)	C1	
			$(E_{\text{photon}} =) \frac{6.63 \times 10^{-34} \times 3.0 \times 10^8}{4.7 \times 10^{-7}}$ or 4.23×10^{-19} (J)	C1	Allow 4.0×10^{-19} (J); which is 2.5 eV
			(number per second = $\frac{0.025}{4.23 \times 10^{-19}}$) number per second = 5.9×10^{16} (s ⁻¹)	A1	Note using 4.0×10^{-19} (J) gives 6.25×10^{16} (s ⁻¹) Examiner's Comments This question required knowledge of both power and energy of photons. It discriminated well with many of the top end candidates getting the correct of 5.9×10^{16} s ⁻¹ . A significant number of candidates scored 1 mark for the energy of the photons. Using the power of 0.025 W in the final step of the calculation proved to be the main obstacle in this calculation. Alternative answers using the energy of a photon as 2.5 eV were allowed. This gave the rate of photons emission to be 4.0×10^{16} s ⁻¹ .
		ii	$(E_{\text{photon}} =) 2.64$ (eV) or $(\phi =) 3.68 \times 10^{-19}$ (J) or $(f_0 =) 5.55 \times 10^{14}$ (Hz) or $(\lambda_0 =) 5.40 \times 10^{-7}$ (m) Photoelectrons are emitted and $2.6(4) > 2.3$ or $4.23 \times 10^{-19} > 3.68 \times 10^{-19}$ or 6.38×10^{14} (Hz) $> 5.55 \times 10^{14}$ (Hz) or 4.7×10^{-7} (m) $< 5.40 \times 10^{-7}$ (m)	M1	Possible ECF from (i) Allow 2.6 (eV) or 3.7×10^{-19} (J) Allow 2.5 (eV) as the energy of the photon
				A1	Note the conclusion must be consistent with (i) Allow $hf > \phi$ Note this can be implied by calculating the KE of the emitted electron



Misconception

The most common mistake was to calculate the energy of the photon in joule, but to write the frequency 6.4×10^{14} on the answer line. This wayward answer can perhaps be explained by frequency and the rate of photon emissions having the same units – s⁻¹.

Question			Answer/Indicative content	Marks	Guidance
					<u>Examiner's Comments</u> Most candidates showed excellent knowledge and understanding of electronvolts and the photoelectric equation. A variety of answers were accepted. The most common approach was to calculate the energy of the photon in eV, and then either show that this was greater than the work function of the metal or to calculate the kinetic energy of the emitted photoelectron. A lot of confidence in the topic of quantum physics was evident in the answers from the candidates. This is illustrated by exemplar 8 below from a middle-grade candidate. Exemplar 8 (ii) The light from the LED is incident on a metal of work function 2.3eV. Explain, with the help of a calculation, whether or not photoelectrons will be emitted from the surface of the metal. $E = \phi + KE_{\max}$ $2.3 \times 1.6 \times 10^{-19} = 3.68 \times 10^{-19} \text{ J}$ $4.23 \times 10^{-19} = 3.68 \times 10^{-19} + KE_{\max}$ photoelectrons will be emitted as the work function ($3.68 \times 10^{-19} \text{ J}$) is less than the energy of a photon ($4.23 \times 10^{-19} \text{ J}$) [2] This exemplar shows the right blend of calculations and scientific text to support the response. Good command of quantum physics earned this candidate full marks.  OCR support Being aware of the contents of the data, formulae and relationship booklet and its layout will support candidates, alleviating the need to recall numerical values of constants and allowing retrieval of correct formulae, or giving assurance that the student has recalled correctly.
			Total	5	

Question	Answer/Indicative content	Marks	Guidance
26	Level 3 (5–6 marks) Description and explanation of pattern changes and quantitatively explains link between de Broglie wavelength and potential difference. <i>There is a well-developed line of reasoning which is clear and logically structured. The information presented is relevant and substantiated.</i> Level 2 (3–4 marks) Clear description of how pattern changes and explanation of pattern changes and qualitatively explains link between de Broglie wavelength and potential difference or limited description of how pattern changes and quantitatively explains link between de Broglie wavelength and potential difference. <i>There is a line of reasoning presented with some structure. The information presented is in the most-part relevant and supported by some evidence.</i> Level 1 (1–2 marks) Limited description of how pattern changes and limited attempts to explain qualitatively the link between de Broglie wavelength and potential difference or qualitatively explains link between de Broglie wavelength and potential difference. <i>The information is basic and communicated in an unstructured way. The information is supported by limited evidence and the relationship to the evidence may not be clear.</i> 0 marks No response or no response worthy of credit.	B1 ×6	Indicative scientific points may include: Description of pattern changes • Rings become closer (not just smaller) • Rings become brighter Qualitative explanation of pattern changes in terms of de Broglie wavelength and potential difference • Electrons gain greater energy • Electrons have a greater speed • Electrons have a greater momentum • Implies smaller wavelength • Smaller wavelength means less diffraction • Shorter wavelength gives shorter path differences between areas of constructive and destructive interference Quantitative explanation of pattern changes in terms of de Broglie wavelength and potential difference • $eV = \frac{1}{2}mv^2$ • $p = mv$ • $v^2 \propto V$ or $p^2 \propto V$ • $\lambda = \frac{h}{p}$ or $\lambda \propto \frac{1}{v}$ • $\lambda = \frac{h}{\sqrt{2meV}}$ or $\lambda \propto \frac{1}{\sqrt{V}}$ <u>Examiner's Comments</u> This question tested an understanding of electron diffraction. Many candidates gave a good qualitative explanation of how the pattern would change. High achieving candidates clearly demonstrated how the de Broglie wavelength λ was related to the potential difference V by equating the energy eV to kinetic energy, then using the definition of momentum and the de Broglie wavelength. Some candidates confused speed v with potential difference V. Many candidates gave a good qualitative explanation. Many candidates did not state that the rings would become brighter.

Question			Answer/Indicative content	Marks	Guidance
					 AfL Candidates should be able to describe how to use light gates. In particular, candidates should be able to indicate the measurements that are needed to determine speed and acceleration. Candidates should state that the light gates should be connected to a timer or data-logger.
					 AfL When analysing experimental data, candidates should be able to determine appropriate graphs to plot which will give a straight line (if the given relationship is true). Candidates should also be able to describe how unknown quantities may be determined using the gradient and / or y-intercept.
					 Misconception There is some confusion between the equations to use for photoelectric effect and the equations to use when considering the de Broglie wavelength. For the de Broglie wavelength, a common misconception is to relate the energy to wavelength by the equation for the energy of a photon, $E = \frac{hc}{\lambda}$
			Total	6	
27			A	1	
			Total	1	
28			$(E =) 1.8 \times 1.6 \times 10^{-19}$ or 2.88×10^{-19} (J)	C1	
			$1.8 \times 1.6 \times 10^{-19} = \frac{6.63 \times 10^{-34} \times 3.0 \times 10^8}{\lambda}$	C1	
			$\lambda = 6.9 \times 10^{-7}$ (m)	A1	
			Total	3	

Question			Answer/Indicative content	Marks	Guidance
29			D	1	
			Total	1	
30		i	$Vq = \frac{1}{2} mv^2$ and $\lambda = \frac{h}{mv}$ Clear algebra leading to $\lambda^2 = \frac{h^2}{2mq} \times \frac{1}{V}$	M1 A1	Allow p for mv Allow e for q in (b)(i) – this is to be treated as a ‘slip’
		ii	1(% uncertainty in λ^2 =) 10% (% uncertainty in λ =) 5% 2Straight line of best fit passes through all error bars 3 gradient = 1.0×10^{-22} $\frac{h^2}{2mq} = \text{gradient}$ $\frac{(6.63 \times 10^{-34})^2}{2 \times m \times 3.2 \times 10^{-19}} = \text{gradient}$ $m = 6.9 \times 10^{-27} \text{ (kg)}$ (hence about 10^{-26} kg)	C1 A1 B1 C1 C1 C1 A1	Note 10 (%) on answer line will score the C1 mark Ignore POT for this mark; Allow $\pm 0.20 \times 10^{-22}$ Possible ECF for incorrect value of gradient Note check for AE (condone rounding error here) and answer must be about 10^{-26} (kg) for any incorrect gradient value for this A1 mark Special case: 1.37×10^{-26} kg scores 3 marks for $q = 1.6 \times 10^{-19}$ C because answer is about 10^{-26} kg
			Total	9	
31			B	1	
			Total	1	
32			$h \rightarrow \text{J s/h} \rightarrow \text{N m s/}$ $\text{J} \rightarrow \text{kg m}^2 \text{ s}^2$ base unit = $\text{kg m}^2 \text{ s}^{-1}$	C1 A1	
			Total	2	